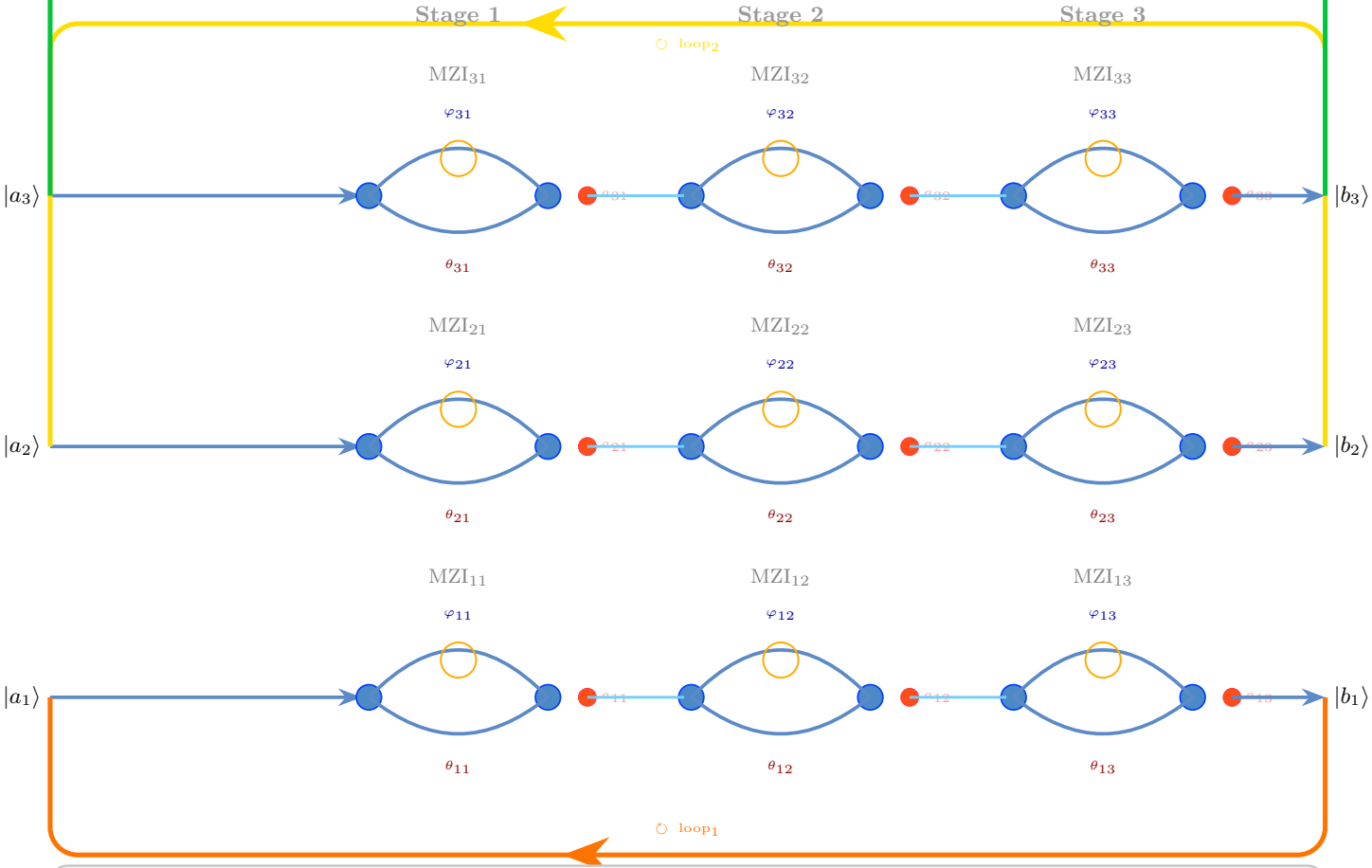


Frobenius Mesh 3×3 | Looping Photons | Complex MZI Matrix



$$A = [a_{ij}]_{3 \times 3}, \quad a_{ij} = \sin(\theta_{ij}) e^{i\varphi_{ij}}, \quad \|A\|_F = \sqrt{\sum_{i,j} |a_{ij}|^2}$$

$A = U \Sigma V^\dagger$ (SVD)

$A^\dagger A \approx I$ (near-unitary)

$\mu \circ \delta = \text{id}$ (IMSCRIB gate)

 beam splitter (BS)

 phase ring φ

 amplitude monitor